

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	18 JUNE 2026
Team ID	INDIVIDUAL PROJECT
Project Name	ShopEZ: E-commerce Application
Maximum Marks	4 Marks

Functional Requirements (FR)

FR No.	Functional Requirement (Epic)	Sub-Requirement (Story / Sub-Task)
FR-1	User Registration	• Registration through secure signup form. • Registration validation using a unique Email ID. • Customer user profile creation (shipping/billing address setups).
FR-2	User Login & Authentication	• Secure login using email and password. • Password recovery module. • User session authentication validation via stateless JSON Web Tokens.
FR-3	Product Discovery & Search	• Instant product catalog browsing via a responsive interface. • High-speed search filtering using single/compound database indices. • Dynamic product category selection (Electronics, Apparel, Home, etc.).
FR-4	Cart & Checkout Submission	• Item cart management (add, edit, or delete items). • Single-page, One-Step secure checkout processing. • Immediate order confirmation receipt and unique Transaction ID generation.
FR-5	Order & Inventory Management	• Admin dashboard view of all active and historical customer orders. • Stock level inventory management (add, edit, update product availability).

FR No.	Functional Requirement (Epic)	Sub-Requirement (Story / Sub-Task)
FR-6	Notification System	• Automatic alerts to users regarding tracking status adjustments. • Real-time delivery fulfillment notifications.

Non-Functional Requirements (NFR)

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The e-commerce application must feature a highly clean, intuitive, and mobile-first responsive layout to prevent interface friction during checkout.
NFR-2	Security	User authentication profiles, account credentials, and transactional checkout parameters must be rigorously protected using JWT access guards and data sanitization.
NFR-3	Reliability	The application must reliably execute single-step checkout workflows and record transaction data accurately without runtime system failures or dropped connections.
NFR-4	Performance	Database search query operations and storefront filtering requests must load efficiently, targeting an optimized runtime latency of under 200 ms .
NFR-5	Availability	The system must be hosted in a resilient staging/web framework environment to ensure constant application uptime for mobile users shopping on the go.
NFR-6	Scalability	The decoupled backend and database structures must allow horizontal scaling to comfortably manage a growing volume of concurrent user traffic, catalogs, and orders.